Resilient Emergency MANETs for Wilderness Safety

One-Page Project Summary

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What this project was. Hikers in New Hampshire's mountains regularly lose cell coverage exactly where emergencies happen. This directed study asked whether a mesh of small, cheap, solar-powered LoRa radios (Meshtastic) could carry emergency traffic — SOS messages, position beacons, short texts — across that terrain, and set out to answer it two ways: build and field-test the data-collection system, and gather ≥2,500 controlled radio measurements to calibrate a terrain-based coverage model.

What happened. The first field trial (Mt. Washington, May) was a systems shakedown: it proved the collection pipeline and showed that free-space coverage assumptions fail badly in real terrain — modeled disagreements up to ∼60 dB on summit-facing links — which redesigned the field protocol around a controlled beacon and direct-link-only data. The project's center of gravity became a purpose-built simulator: two independent implementations (Python and Rust) of the same statewide mesh model, cross-checked against each other, that simulate a full year of network life — radios, batteries, solar, weather, hikers, emergencies — in about 3.5 minutes. Its principal finding (a model result awaiting field calibration): *node survival is governed by how much time radios spend listening, not by which routing algorithm they run.* Five different routing algorithms differed by less than 0.3% in annual battery depletions, while duty-cycling policies (letting radios sleep between listen windows) cut depletions 5.5–24× — at a quantified cost in delivery rate and SOS latency, so the result is a design trade, not a free win. The second field campaign (July) was re-planned in flight after the beacon radio's hardware failed — its battery, then its charging port — so the calibration dataset was not collected (0 of the 2,500 points). The final field day (Pack Monadnock) still ran the receiver system end-to-end and measured exactly why public mesh stations cannot substitute for a controlled beacon: they transmit only 0–4 times per hour, and every packet received arrived through relays, making signal strength unattributable to any path. Every change of plan was registered, with cryptographic hashes, before the data it affected was collected.

What it is now. A working, documented, reproducible research platform: the two-engine simulator with a released, hash-locked results set; a terrain-aware (Longley–Rice) planning layer; a preregistered field protocol with frozen prediction packs ready to score; a hardened field acquisition rig proven in the field; and a complete evidence chain — every number in the final report traces to a checksummed artifact in the repository (`https://github.com/edoherty2017/resilient-emergency-manets`). One small step remains to close the original calibration goal: a ∼$25 replacement beacon board and a single two-radio field day under the already-frozen protocol. Full detail: *Directed Study Final Report*, 2026-07-26.